

International Institute of Information Technology, Hyderabad
(Deemed to be University)

CS1.302 - Automata Theory - Monsoon 2026

Final Exam

Maximum Time : 90 Minutes

Total Marks : 35

Roll No. _____

Programme _____

Date _____

Room No. _____

Seat No. _____

Invigilator Sign. _____

Do not write here	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Bonus
Marks Scored										
Total Marks	2	4	4	3	4	6	2	3	7	
Post Paper Showing										
TA Sign										

General Instructions to the students

1. Place your Permanent / Temporary Student ID card on the desk during the examination for verification by the Invigilator.
2. **Do not answer with pencil. Mark the choice clearly.**
3. **No questions will be answered during the exam. Make necessary and reasonable assumptions.**
4. **No extra sheets will be provided.**
5. Reading material such as books (unless open book exam) are not allowed inside the examination hall.
6. Borrowing writing material or calculators from other students in the examination hall is prohibited.
7. **If any student is found indulging in malpractice or copying in the examination hall, the student will be given 'F' grade for the course and may be debarred from writing other examinations.**
8. If you write the answer elsewhere on the booklet, mention it below the question.

Standard Notation Followed in the Questions

1. FSM stands for finite state machine.
2. DFA stands for deterministic finite automata.
3. NFA stands for non-deterministic finite automata.
4. PDA stands for Push Down Automata.
5. CFG stands for context-free grammar.
6. a^* is the Kleene Star operation.
7. Φ denotes the empty language.
8. ϵ denotes the empty string.

Write detailed answers. Adequately explain your assumptions and thought process.

Best of Luck

1. [2 points] Describe the language generated by this grammar:

- $S \longrightarrow 0S \mid 1A$
- $A \longrightarrow 0A \mid 1B$
- $B \longrightarrow 0B \mid 1B \mid \epsilon$

[CO1, CO2, CO3]

2. [4 points] For a string $w \in \{a, b\}^*$, let $\#_a(w)$ denote the number of a 's in w and $\#_b(w)$ the number of b 's in w . Consider

$$L = \{ w \in \{a, b\}^* \mid \#_a(w) = 2 \#_b(w) \}.$$

Construct a pushdown automaton accepting L .

(Hint: While aab is straightforward, you need a way to accept strings like $ababaa$ where the a 's read so far is less than twice the number of b 's. Construct the PDA only. No accompanying explanation is required.) [CO1, CO2]

3. [4 points] Consider the following language:

$$\text{EQUIV}_{TM} := \{ \langle M_1, M_2 \rangle \mid M_1, M_2 \text{ are TMs and } L(M_1) = L(M_2) \}$$

Prove that EQUIV_{TM} is undecidable by reducing HALT to it.

Hint: Try thinking about the language computed by a machine that accepts if and only if a TM M run on string w halts. [CO3, CO4]

4. [3 points] Let $G = (V, \Sigma, R, S)$ be a Context-Free Grammar in Chomsky Normal Form having exactly k variables ($|V| = k$).

Prove that if there exists some string $w \in L(G)$ whose derivation requires at least 2^k steps, then $L(G)$ must be an infinite language.

[CO1, CO3]

5. [4 points] Let $DFA(q)$ denote the set of languages L over the alphabet $\Sigma = \{a\}$, that have a DFA with $\leq q$ states that accepts L . Observe that $DFA(q) \subseteq DFA(q+1)$ for $q > 0$. Prove that

$$DFA(q) \neq DFA(q+1)$$

for all $q > 0$, by showing the existence of a language L that has a DFA with $q+1$ states that decides L , yet has no DFA of q states that decides L .

Hint: Consider the language L which only accepts strings of length $0 \pmod{q+1}$, and show it has a DFA of $q+1$ states. Try using the pigeonhole principle to show $L \notin DFA(q)$.

[CO4]

6. [6 points] **(Turing's incomputable number)** Recall that as the set of all Turing Machines have a bijection to the set of finite binary strings, we can thus enumerate them as $T_1, T_2, \dots$ where $T_i = i$ written in base 2. Define the base-2 real number $\chi = 0.\chi_1\chi_2\dots$ where $\chi_i = 1$ if and only if T_i on input $\langle i \rangle$ halts, else $\chi_i = 0$.
- a) [1 point] Show that χ is uncomputable; there does not exist a Turing Machine that when queried any $n \in \mathbb{N}$ returns the n^{th} bit of χ .
- b) [5 points] Show that χ is however *lower semi-computable*; there is a deterministic algorithm which, when given an input $n \in \mathbb{N}$, returns a base-2 real number χ'_n , such that $\chi'_n \leq 0.\chi_1\dots\chi_n$, and $\lim_{n \rightarrow \infty} \chi'_n = \chi$.
(Hint: Use dovetailing, and that the enumeration is in increasing order of size.)

[CO1, CO4]

7. [2 points] What is the complement of the regular language $(aaa)^*$? Write its regular expression. Assume $\Sigma = \{a\}$. [CO1, CO3]

8. [3 points] Let R_1, R_2 be two recursive languages, and RE_1, RE_2 be two recursively enumerable languages. Classify the following languages with **proper justification**:

- $R_1 \cup R_2$
- $RE_1 \cup RE_2$
- $RE_1 \cap R_2$

[CO1, CO3]

9. [7 points] A n -state *Busy Beaver* is a deterministic Turing Machine with n **non-rejecting and non-accepting** states, defined on a single two-way infinite tape, such that, when the initial configuration of the tape is such that every cell has just 0, it eventually halts (see below image for an example of a 2-state Busy Beaver (the accepting state is not counted as per the definition) that halts in 3 steps on the all-zero tape.). Note that the tape head can only move left or right in a transition, it cannot stay in place. The *Busy Beaver number* $BB(n)$ is the maximum number of steps any n -state Busy Beaver takes before halting.

Show that $BB(2) \geq 6$ by constructing a 2-state Busy Beaver that halts in 6 steps (in fact it can be shown that $BB(2) = 6$). You need only provide the state diagram, and you can omit transitions that won't be taken in the run on the all-zero tape (as in the example below).

Partials shall be awarded as follows: for providing a valid 2-state Busy Beaver that halts in 4 steps you will receive 2 marks, for 5 steps 4 marks, and for 6 steps full marks.

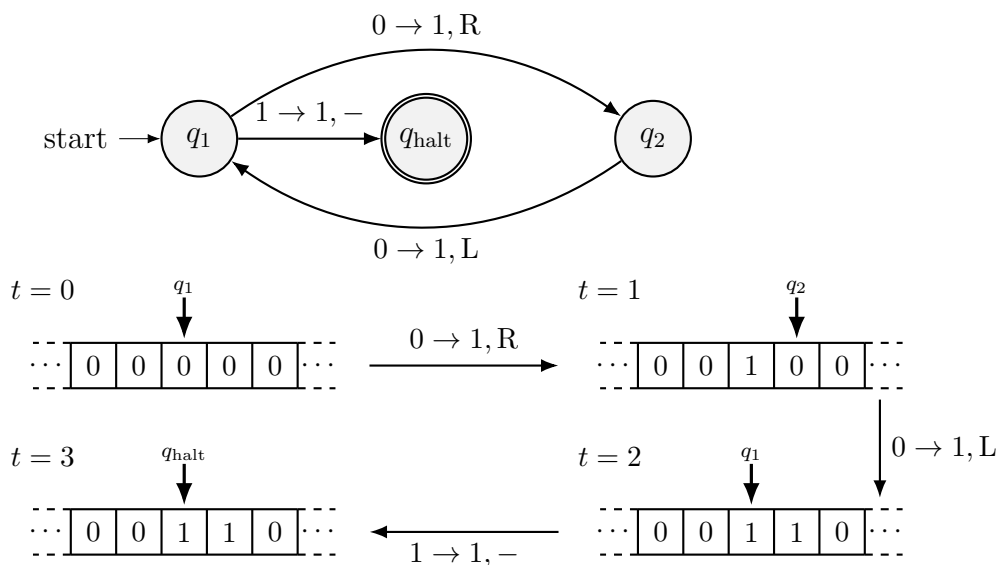


Figure 1: State transition diagram and execution of a 2-state Busy Beaver. The machine starts in q_1 with an all-zero tape and halts after three transitions. You can assume the missing transitions all just lead to a rejecting state.

[CO1, CO2, CO3, CO4]

BONUS (Non-mandatory Bonus Question)

Recall the definition of a group $(G, \cdot)$ as some set of elements G with an operation $(\cdot) : G \times G \mapsto G$ such that:

- **Closure:** $a \cdot b \in G$ for all $a, b \in G$.
- **Associativity:** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in G$.
- **Identity:** There exists a unique identity e_G such that $a \cdot e_G = a$ for all $a \in G$.
- **Inverse:** For all $a \in G$ there exists a unique inverse a^{-1} such that $a \cdot a^{-1} = e_G$.

Recall that a subgroup H of G is a subset $H \subseteq G$ such that the above axioms hold for elements in H (you can use without proof that $e_G = e_H \in H$). For any $a \in G$, the *right coset* Ha of a subgroup H is the set $Ha = \{ha \mid h \in H\}$.

For a language $L \subseteq \Sigma^*$, we say two strings $x, y \in \Sigma^*$ are equivalent under L , or $x \equiv_L y$, if for all $z \in \Sigma^*$, we have $xz \in L$ if and only if $yz \in L$. Let $(G, \cdot)$ be a finite group ($|G|$ is finite), and let $\Sigma = G$ be the alphabet. For a string of elements $s = x_1x_2 \dots x_n \in \Sigma^n$, define $\phi(s)$ to be the evaluation of the group operation on the elements of s , hence $\phi(s) = x_1 \cdot x_2 \cdots x_n$. Note that if $s = \varepsilon$ then $\phi(s) = e$.

For any subgroup H of G define L_H as the set of words that evaluate to an element in H , i.e

$$L(H) = \{s \in \Sigma^* \mid \phi(s) \in H\}$$

1. Prove that for any $x, y \in \Sigma^*$ $x \equiv_{L_H} y$ if and only if $\phi(x)$ and $\phi(y)$ belong to the same right coset Ha of H , where $a \in G$.
(Hint: You may use the following version of the Myhill-Nerode Theorem without proof: For a regular language L , $x \equiv_L y$ if and only if $xz \in L \iff yz \in L$ for all $z \in \Sigma^*$).
2. Thus, deduce that the minimal DFA that accepts L_H has $\frac{|G|}{|H|}$ states.
(Hint: You can use without proof the fact that the number of distinct right cosets Hx of a subgroup H of G is $\frac{|G|}{|H|}$; this is implied by Lagrange's Theorem).

[CO1, CO3]

N.B: Answering a bonus question is not mandatory. Correct answers to the bonus question will fetch you extra points. This will be added to the total score you get in the course as a “top-up”.

Rough Work

